THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 5: **Life Processes** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A heart pumps 70 mL of blood per beat at 72 beats per minute. Volume pumped in a minute (in mL) is: 1

(a) 720 **(b)** 7200 **(c)** 504 **(d)** 5040

2. Lime water turns milky faster in a test tube with exhaled air than in inhaled air because exhaled air has: 1

(a) more oxygen **(b)** less water

(c) more carbon dioxide **(d)** more nitrogen

3. Assertion (A): Oxygenated and deoxygenated blood are not allowed to mix in human heart. Reason (R): This allows a more efficient supply of oxygen to the body. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

4. Haemoglobin is present in the: 1

(a) white blood cells **(b)** plasma

(c) red blood cells **(d)** platelets

5. A man's blood sugar leaks into urine. This indicates a problem in: 1

(a) reabsorption in nephron **(b)** lungs

(c) filtration in glomerulus **(d)** stomach

6. Assertion (A): Alveoli have thin walls and rich capillary network. Reason (R): This allows rapid gaseous exchange. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

7. A person's glomerular filtrate is 180 L per day, and urine excreted is 1.8 L. The percentage of water reabsorbed is: 1

(a) 50% **(b)** 99% **(c)** 95% **(d)** 90%

8. A student finds the pH of mouth falling after sugar meals. The relation of digestion: 1

(a) saliva becomes acidic **(b)** amylase is acidic

(c) starch digests **(d)** bacteria produce acid from sugar

9. **Assertion (A):** In yeast, pyruvate is converted into ethanol and carbon dioxide in the absence of oxygen. 1
Reason (R): Anaerobic respiration releases more energy than aerobic respiration.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Xylem carries the food prepared in the leaves to the other parts of the plant. 1
Reason (R): Translocation of food takes place through the phloem using energy.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A plant is kept in a closed bell jar with NaOH solution. Why does it fail to photosynthesise? 2

12. A thin slice of ring (bark) is removed from a tree trunk. What happens to the roots and why? 2

13. Why is photosynthesis called the process that sustains all life on Earth? 2

14. What would happen if the guard cells of a plant could not open? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain the importance of the length of the small intestine in herbivores and carnivores with reasons. 3

16. Why does the rate of transpiration increase on a hot, windy day and what is its advantage? 3

17. Compare nutrition in Cuscuta and green plant. 3

OR

Explain why blood flows only in one direction in the circulatory system. 3

18. Differentiate between photosynthesis and respiration (three points). 3

OR

What are the adaptations in the nephron and the glomerulus for efficient excretion? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** During a health check-up, a doctor measured the blood pressure of Mr Rao's grandson Arjun and recorded 120/80 mm of Hg. She explained that the human heart has four chambers and that oxygen-rich blood and oxygen-poor blood do not mix. 5
(i) How many chambers are there in the human heart?
(ii) Name the vessel that brings oxygenated blood from the lungs to the heart.
(iii) Name the instrument used to measure blood pressure.
(iv) Why is it necessary to separate oxygenated and deoxygenated blood in birds and mammals?

20. Compare the circulatory system of fish and human beings with the pathway of blood and the reason for the difference. 5

OR

Compare the working of the heart in a person at rest and during exercise: heart rate, stroke volume, breathing, and the muscle metabolism. Explain why lactic acid may accumulate. 5

END OF QUESTION PAPER — Total: 40 Marks